

Shaft								
MDESIGN 2020 - Second Edition								
Producer:	Project:							
Verified by:	Date: 27.04.2026							
Customer: Student								
This program allows to prove the bearing ability for shafts and axles. The calculation base is provided by DIN 743, edition of December 2012. The proof of the bearing ability for shafts an axles is produced by defining a calculated safety. This safety is divided in the safety against fatigue fracture and the residual deformation (and flaw or forced break). When calculating the avoidance of fatigue fracture, constant stress amplitudes being equivalent to damaging loads are taken as a basis. These ones are resulting from the predetermined loads. When proving against the residual deformation or forced break, designated as a safety against yielding, only the maximum occurring load is determinant. This one is resulting from the predetermined loads, too. The calculation of factors of safety is related only to the point of a clear notch effect. For it, 9 calculable notches are at your disposal due to the graphical selection, principally. The scope is limited to steels. Welded members should be calculated separately. But the utilized standard or the present program is ineffective for this purpose! The calculation base for the module Shaft Calculation is provided by DIN 743, edition of December 2012, part 1-4 "Tragfähigkeitsberechnung von Wellen and Achsen" ("Calculation of bearing capacity of shafts and axles").								
<u>Input data:</u> Shaft calculation in accordance with DIN 743 - extended version <table border="0" style="width: 100%;"> <tr> <td style="width: 50%;">Geometry scheme</td> <td style="width: 50%;">General shaft geometry</td> </tr> <tr> <td>Calculation process</td> <td>Dynamic and static strength proof</td> </tr> <tr> <td colspan="2"> Geometry Shaft geometry </td> </tr> </table>			Geometry scheme	General shaft geometry	Calculation process	Dynamic and static strength proof	Geometry Shaft geometry	
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Last updated: 13.05.2026		Page: 1 / 31						

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Nr.	D _{al} mm	D _{il} mm	D _{ar} mm	D _{ir} mm	L mm	R _z μm	r mm	d: mm	t: mm	α _σ zd:	α _{σb} :	α _{τt} :	n _{zd} :	n _b :	n _t :	β _σ zddBK :	β _σ bdBK :	β _τ dBK: :	d _{BK} :																																																																																																																																																														
1	22	0	22	0	50	1,6	1	0	0	0	0	0	0	0	0	0	0	0	0																																																																																																																																																														
2	24	0	24	0	24	1,6	0,8	0	0	0	0	0	0	0	0	0	0	0	0																																																																																																																																																														
3	28	0	28	0	20	1,6	1	0	0	0	0	0	0	0	0	0	0	0	0																																																																																																																																																														
4	30,048	0	30,048	0	31	1,6	1	0	0	0	0	0	0	0	0	0	0	0	0																																																																																																																																																														
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Last updated: 13.05.2026										Module version: 18.0.13e					Page: 2 / 31																																																																																																																																																																		

Shaft

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Project:

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Type of loading: tension-pressure

Dynamically pure cyclic

Type of loading: bending

Dynamically pure cyclic

Type of loading: torsion

Dynamically pure cyclic

Factor for maximum loading (tension-pressure)

1

Factor for maximum loading (bending)

1

Factor for maximum loading (torsion)

1

Axial forces F_{ax}

Nr.	Position x = mm	Amount = N	Radius = mm	Angle α = °
1	109,5	-694,42	15,024	0

Radial forces F_r

Nr.	Position x = mm	Amount = N	Angle α = °
1	20	-1213,9421	270
2	109,5	-1213,9421	90
3	109,5	-2785	180

Torsion

Nr.	Position x = mm	Torsion moments M_t : N·mm	Power P: kW	Transition part =
1	20	55704	0	drive
2	109,5	55704	0	takeoff

Specifications about the load/loadings

Shaft					
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Customer: Student					
Loading case Calculation of finite-life fatigue strength ? Load cycles until fatigue strength Required load cycles Slope exponent of S-N curve normal stress Slope exponent of S-N curve shear stress Minimum safety against fatigue fracture Minimum safety against residual deformation Minimum safety against incipient crack with hard surface		Constant mean stress (loading case 1) yes $N_D = 1000000$ $N_L = 10000000$ $q_\sigma : 5$ $q_\tau : 8$ $S_{Dmin} = 1,2$ $S_{Fmin} = 1,2$ $S_{Gmin} = 1,2$			
Material Data Strength values according to Material designation Material number Gage diameter For the gage diameter Tensile strength Yield stress Cyclic fatigue strength under bending stress Cyclic tension and pressure fatigue strength Cyclic torsional fatigue strength Young's modulus Shear modulus Density Apply surface hardening to Material group Heat treatment Surface hardening				Own assignment 20TiMnCrS4 1.7323 $d_B = 16$ mm $\sigma_{B'} (R_m) = 1100$ N/mm ² $\sigma_{S'} (R_e) = 850$ N/mm ² $\sigma_{bW'} = 330$ N/mm ² $\sigma_{zdW'} = 150$ N/mm ² $\tau_{tW'} = 90$ N/mm ² $E = 215000$ N/mm ² $G = 83000$ N/mm ² $\rho = 7850$ kg/m ³ Total shaft Cemented steels trial hardened no	
Last updated: 13.05.2026		Module version: 18.0.13e			
Page: 4 / 31					

Shaft

LOGO

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Producer:

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Results:

Calculation process:

Dynamic and static strength proof

Geometry

Total shaft length	L	=	155	mm
Total shaft mass	m	=	0,623	kg
Mass moment of inertia of the shaft	J	=	0,00005	kg·m ²
Geometrical moment of inertia of the shaft	I	=	14,443	cm ⁴
Position of the centre of gravity in the X-axis	x _s	=	84,447	mm
Angle of torsion	φ	=	0,099	°

Additional shaft data:

Shaft fillet number	l mm	I _p cm ⁴	W _t cm ³	m kg	J kg·m ²	I cm ⁴	W _b cm ³
1	50	2,3	2,091	0,149	0	1,15	1,045
2	24	3,257	2,714	0,085	0	1,629	1,357
3	20	6,034	4,31	0,097	0	3,017	2,155
4	31	8,003	5,327	0,173	0	4,002	2,663
5	10	6,034	4,31	0,048	0	3,017	2,155
6	20	3,257	2,714	0,071	0	1,629	1,357

Loading Data

Calculation results for point	x	=	142	mm
Trend of curve of the transverse force	Q _x	=	1508,998	N
deflection	y _x	=	0	mm
Angle of deflection	Θ	=	0,011259	°

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Supporting forces: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 10px;"> <thead> <tr> <th style="width: 5%;">No.</th> <th style="width: 15%;">Type</th> <th style="width: 5%;">Position x mm</th> <th style="width: 10%;">Radial force in the Y-axis R_y N</th> <th style="width: 10%;">Radial force in the Z-axis R_z N</th> <th style="width: 10%;">Result. radial force R N</th> <th style="width: 10%;">Axial force in the X-axis R_{ax} N</th> <th style="width: 10%;">Tilting moment in the Y-axis N·m</th> <th style="width: 10%;">Tilting moment in the Z-axis N·m</th> <th style="width: 10%;">Result. tilting moment N·m</th> </tr> </thead> <tbody> <tr> <td style="text-align: center;">1</td> <td>Locating bearing -></td> <td style="text-align: center;">70</td> <td style="text-align: center;">-1107,332</td> <td style="text-align: center;">-1508,997</td> <td style="text-align: center;">1871,699</td> <td style="text-align: center;">694,42</td> <td style="text-align: center;">0</td> <td style="text-align: center;">0</td> <td style="text-align: center;">0</td> </tr> <tr> <td style="text-align: center;">2</td> <td>Locating bearing <-</td> <td style="text-align: center;">142</td> <td style="text-align: center;">-1671,558</td> <td style="text-align: center;">1508,997</td> <td style="text-align: center;">2251,928</td> <td style="text-align: center;">0</td> <td style="text-align: center;">0</td> <td style="text-align: center;">0</td> <td style="text-align: center;">0</td> </tr> </tbody> </table>										No.	Type	Position x mm	Radial force in the Y-axis R_y N	Radial force in the Z-axis R_z N	Result. radial force R N	Axial force in the X-axis R_{ax} N	Tilting moment in the Y-axis N·m	Tilting moment in the Z-axis N·m	Result. tilting moment N·m	1	Locating bearing ->	70	-1107,332	-1508,997	1871,699	694,42	0	0	0	2	Locating bearing <-	142	-1671,558	1508,997	2251,928	0	0	0	0
No.	Type	Position x mm	Radial force in the Y-axis R_y N	Radial force in the Z-axis R_z N	Result. radial force R N	Axial force in the X-axis R_{ax} N	Tilting moment in the Y-axis N·m	Tilting moment in the Z-axis N·m	Result. tilting moment N·m																														
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2	Locating bearing <-	142	-1671,558	1508,997	2251,928	0	0	0	0																														
Resulting maximum bending moment: Position x = 109,5 mm Amount M_{bmax} = 73,218 N·m Resulting maximum torsional moment: Position x = 20 mm Amount M_{tmax} = 55,704 N·m Resulting maximum tension-pressure-force: Position x = 70 mm Amount F_{zmax} = -694,42 N Resulting maximum tension-pressure-stress: Position x = 70 mm Amount σ_{zmax} = -1,535 N/mm² Resulting maximum bending stress: Position x = 70 mm Amount σ_{bmax} = 44,723 N/mm² Resulting maximum torsional stress: Position x = 20 mm Amount τ_{tmax} = 26,643 N/mm² Resulting maximum equivalent stress: Position x = 49,821 mm Amount σ_{vmax} = 57,821 N/mm² Resulting maximum deflection:																																							
Last updated: 13.05.2026					Module version: 18.0.13e			Page: 6 / 31																															

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Position x = 0 mm Amount y_{max} = 0,046557 mm Angle of the maximum deflection: Position x = 3,986 mm Amount Θ = 0,045533 ° Minimum safety against yielding: Position x = 70 mm Amount S_F = 17,412 Minimum safety against fatigue fracture: Position x = 73,514 mm Amount S_D = 1,98 Minimum safety against incipient crack with hard surface: Position x = 73,514 mm Amount S_G = 22,265 Parameter of cross-sections: Tension-pressure force F_{zd} and tension/pressure stress σ_{zd} <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 10px;"> <thead> <tr> <th>No.</th> <th>Type</th> <th>Position x mm</th> <th>Result. F_{zdx} N</th> <th>Amplitude F_{zda} N</th> <th>Mean F_{zdm} N</th> <th>Maximum F_{zdmax} N</th> <th>Amplitude σ_{zda} N/mm²</th> <th>Mean σ_{zdm} N/mm²</th> <th>Maximum σ_{zdmax} N/mm²</th> </tr> </thead> <tbody> <tr><td>1</td><td>Shaft fillet</td><td>50</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>2</td><td>Shaft fillet</td><td>74</td><td>-694,42</td><td>-694,42</td><td>0</td><td>-694,42</td><td>-1,535</td><td>0</td><td>-1,535</td></tr> <tr><td>3</td><td>Shaft fillet</td><td>94</td><td>-694,42</td><td>-694,42</td><td>0</td><td>-694,42</td><td>-1,128</td><td>0</td><td>-1,128</td></tr> <tr><td>4</td><td>Shaft fillet</td><td>125</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>5</td><td>Shaft fillet</td><td>135</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>6</td><td>Calculation results for point x</td><td>142</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> </tbody> </table>				No.	Type	Position x mm	Result. F_{zdx} N	Amplitude F_{zda} N	Mean F_{zdm} N	Maximum F_{zdmax} N	Amplitude σ_{zda} N/mm ²	Mean σ_{zdm} N/mm ²	Maximum σ_{zdmax} N/mm ²	1	Shaft fillet	50	0	0	0	0	0	0	0	2	Shaft fillet	74	-694,42	-694,42	0	-694,42	-1,535	0	-1,535	3	Shaft fillet	94	-694,42	-694,42	0	-694,42	-1,128	0	-1,128	4	Shaft fillet	125	0	0	0	0	0	0	0	5	Shaft fillet	135	0	0	0	0	0	0	0	6	Calculation results for point x	142	0	0	0	0	0	0	0
No.	Type	Position x mm	Result. F_{zdx} N	Amplitude F_{zda} N	Mean F_{zdm} N	Maximum F_{zdmax} N	Amplitude σ_{zda} N/mm ²	Mean σ_{zdm} N/mm ²	Maximum σ_{zdmax} N/mm ²																																																																
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Last updated: 13.05.2026		Module version: 18.0.13e																																																																							
		Page: 7 / 31																																																																							

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Bending moment M_b and bending stress σ_b

No.	Type	Position x mm	Result. M_{bx} N·m	Amplitude M_{ba} N·m	Mean M_{bm} N·m	Maximum M_{bmax} N·m	Amplitude σ_{ba} N/mm ²	Mean σ_{bm} N/mm ²	Maximum σ_{bmax} N/mm ²
1	Shaft fillet	50	36,418	36,418	0	36,418	34,838	0	34,838
2	Shaft fillet	74	59,688	59,688	0	59,688	43,98	0	43,98
3	Shaft fillet	94	59,902	59,902	0	59,902	27,795	0	27,795
4	Shaft fillet	125	38,295	38,295	0	38,295	17,769	0	17,769
5	Shaft fillet	135	15,769	15,769	0	15,769	11,619	0	11,619
6	Calculation results for point x	142	0,003	0,003	0	0,003	0,002	0	0,002

Torsional moment M_t und Torsional stress τ_t

No.	Type	Position x mm	Result. M_{tx} N·m	Amplitude M_{ta} N·m	Mean M_{tm} N·m	Maximum M_{tmax} N·m	Amplitude τ_{ta} N/mm ²	Mean τ_{tm} N/mm ²	Maximum τ_{tmax} N/mm ²
1	Shaft fillet	50	55,704	55,704	0	55,704	26,643	0	26,643
2	Shaft fillet	74	55,704	55,704	0	55,704	20,522	0	20,522
3	Shaft fillet	94	55,704	55,704	0	55,704	12,924	0	12,924
4	Shaft fillet	125	0	0	0	0	0	0	0
5	Shaft fillet	135	0	0	0	0	0	0	0
6	Calculation results for point x	142	0	0	0	0	0	0	0

Critical shaft speed values:

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Critical bending shaft speed values		
No.	Critical shaft speed values n_b 1/min	Eigenfrequencies ω rad/s
1	160803,76	16839,33
2	621861,96	65121,23
3	915163,3	95835,68
4	1733207,75	181501,09
5	2105051,63	220440,49
Critical torsional shaft speed values		
No.	Critical shaft speed values n_b 1/min	Eigenfrequencies ω rad/s
1	704614,64	73787,07
2	1362603,12	142691,47
3	1924384,22	201521,04
4	2488904,31	260637,45
5	3112602,48	325950,97
Material Data		
Material parameter for	$d_{\max}$	= 30,048 mm
Material designation		20TiMnCrS4
Material number		1.7323
Tensile strength	σ_B	= 1100 N/mm ²
Yield stress	σ_S	= 850 N/mm ²
Cyclic tension and pressure fatigue strength	σ_{zdW}	= 150 N/mm ²
Cyclic fatigue strength under bending stress	σ_{bW}	= 330 N/mm ²
Cyclic torsional fatigue strength	τ_{tW}	= 90 N/mm ²
Last updated: 13.05.2026		Module version: 18.0.13e
		Page: 9 / 31

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Technological dimension factor
(tensile strength)

$$K_{1B}(d_{\max}) = 1$$

Technological dimension factor
(yield stress)

$$K_{1S}(d_{\max}) = 1$$

Proof of Strength

$K_1(d)$ - Technological dimension factor

$K_2(d)$ - Geometrical dimension factor

K_F - Influence factor of surface roughness

$\alpha_{\sigma, \tau}$ - Form factors

No.	Type	Position x mm	K_{1B} (d)	K_{1S} (d)	Tension - pressure $K_2(d)$	Bending and torsion $K_2(d)$	Tension- pressure, bending $K_{F\sigma}$	Torsion $K_{F\tau}$	Tension - pressure $\alpha_{\sigma zd}$	Bending $\alpha_{\sigma b}$	Torsion α_{τ}
1	Shaft fillet	50	1	1	1	0,93	0,97	0,98	2	1,84	1,4
2	Shaft fillet	74	1	1	1	0,92	0,97	0,98	2,4	2,2	1,58
3	Shaft fillet	94	1	1	1	0,91	0,97	0,98	2,06	1,89	1,42
4	Shaft fillet	125	1	1	1	0,91	0,97	0,98	2,06	1,89	1,42
5	Shaft fillet	135	1	1	1	0,92	0,97	0,98	2,4	2,2	1,58
6	Calculation results for point x	142	1	1	1	0,92	0,97	0,98	-	-	-

Shaft

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 G' - Relative stress drop $n_{\sigma, \tau}$ - Bearing factor

No.	Type	Position x mm	Tension- pressure G'_{zd} 1/mm	Bending G'_b 1/mm	Torsion G'_t 1/mm	Tension- pressure $n_{\sigma zd}$	Bending n_{ob}	Torsion n_{τ}
1	Shaft fillet	50	2,68	2,68	1,15	1,05	1,05	1,03
2	Shaft fillet	74	3,22	3,22	1,44	1,05	1,05	1,04
3	Shaft fillet	94	2,68	2,68	1,15	1,05	1,05	1,03
4	Shaft fillet	125	2,68	2,68	1,15	1,05	1,05	1,03
5	Shaft fillet	135	3,22	3,22	1,44	1,05	1,05	1,04
6	Calculation results for point x	142	-	-	-	-	-	-

 $\beta_{\sigma ddBK}, \beta_{obdBK}, \beta_{\tau dBK}$ - Stress concentration factor at d_{BK} $\beta_{\sigma zd}, \beta_{ob}, \beta_{\tau}$ - Stress concentration factors K_v - Influence factor of surface hardening

No.	Type	Position x mm	Tension - pressure $\beta_{\sigma ddBK}$	Bending g β_{obdBK}	Torsion $\beta_{\tau dBK}$	Tension - pressur e $\beta_{\sigma zd}$	Bending β_{ob}	Torsion β_{τ}	Tension - pressur e K_{vzd}	Bending g K_{vb}	Torsion $K_{v\tau}$
1	Shaft fillet	50	-	-	-	1,91	1,75	1,35	1	1	1
2	Shaft fillet	74	-	-	-	2,27	2,08	1,53	1	1	1
3	Shaft fillet	94	-	-	-	1,96	1,8	1,37	1	1	1
4	Shaft fillet	125	-	-	-	1,96	1,8	1,37	1	1	1
5	Shaft fillet	135	-	-	-	2,27	2,08	1,53	1	1	1
6	Calculation results for point x	142	-	-	-	1	1	1	1	1	1

Shaft


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 K_σ, K_τ - Total influence factor $\sigma_{zdWK}, \sigma_{bWK}, \tau_{tWK}$ - Cyclic fatigue strength of the notched part K_{2F} - Static bearing effect

No.	Type	Position x mm	Tension - pressur e K_σ	Bending K_σ	Torsion K_τ	Tension - pressur e σ_{zdWK} N/mm ²	Bending σ_{bWK} N/mm ²	Torsions τ_{tWK} N/mm ²	Tension - pressur e K_{2Fzd}	Bendin g K_{2Fb}	Torsion K_{2Ft}
1	Shaft fillet	50	1,94	1,92	1,48	77,26	171,87	60,94	1	1,2	1,2
2	Shaft fillet	74	2,31	2,29	1,68	65	143,85	53,68	1	1,2	1,2
3	Shaft fillet	94	2	2,01	1,52	75,12	163,96	59,05	1	1,2	1,2
4	Shaft fillet	125	2	2,01	1,52	75,12	163,96	59,05	1	1,2	1,2
5	Shaft fillet	135	2,31	2,29	1,68	65	143,85	53,68	1	1,2	1,2
6	Calculation results for point x	142	1,03	1,12	1,1	145,01	295,02	81,55	1	1,2	1,2

 γ_F - Yield point rise $\sigma_{zdFK}, \sigma_{bFK}, \tau_{tFK}$ - Yield point of the part

No.	Type	Position x mm	Tension- pressure γ_{Fzd}	Bending γ_{Fb}	Torsion γ_{Ft}	Tension- pressure σ_{zdFK} N/mm ²	Bending σ_{bFK} N/mm ²	Torsion τ_{tFK} N/mm ²
1	Shaft fillet	50	1,1	1,05	1	935	1071	588,9
2	Shaft fillet	74	1,1	1,1	1	935	1122	588,9
3	Shaft fillet	94	1,1	1,05	1	935	1071	588,9
4	Shaft fillet	125	1,1	1,05	1	935	1071	588,9
5	Shaft fillet	135	1,1	1,1	1	935	1122	588,9
6	Calculation results for point x	142	1	1	1	850	1020	588,9

Shaft


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Static safety

No.	Type	Position x mm	S_F	In Point1 S_{F1}	in Point2 S_{F2}
1	Shaft fillet	50	17,95	-	-
2	Shaft fillet	74	18,63	-	-
3	Shaft fillet	94	28,64	-	-
4	Shaft fillet	125	60,27	-	-
5	Shaft fillet	135	96,57	-	-
6	Calculation results for point x	142	4,7e+005	-	-

 ψ - Influence factor of the mean stress sensitivtz σ_{mv} , τ_{mv} - Comparative mean stress

No.	Type	Position x mm	Tension - pressure $\psi_{zd\sigma K}$	Bending $\psi_{b\sigma K}$	Torsion $\psi_{\tau K}$	σ_{mv} N/mm ²	τ_{mv} N/mm ²	σ_{mv1} N/mm ²	τ_{mv1} N/mm ²	σ_{mv2} N/mm ²	τ_{mv2} N/mm ²
1	Shaft fillet	50	-	0,08	0,03	0	0	-	-	-	-
2	Shaft fillet	74	0,03	0,07	0,03	0	0	-	-	-	-
3	Shaft fillet	94	0,04	0,08	0,03	0	0	-	-	-	-
4	Shaft fillet	125	-	0,08	-	0	0	-	-	-	-
5	Shaft fillet	135	-	0,07	-	0	0	-	-	-	-
6	Calculation results for point x	142	-	0,15	-	0	0	-	-	-	-

Shaft											
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Alternating fatigue strength of the part (rated fatigue limit)											
No.	Type	Position x mm	Tension - pressure σ_{zADK} N/mm ²	Bending σ_{bADK} N/mm ²	Torsion τ_{tADK} N/mm ²	Tension - pressure in Point1 σ_{zADK1} N/mm ²	Bending in Point1 σ_{bADK1} N/mm ²	Torsion in Point1 τ_{tADK1} N/mm ²	Tension - pressure in Point2 σ_{zADK2} N/mm ²	Bending in Point2 σ_{bADK2} N/mm ²	Torsion in Point2 τ_{tADK2} N/mm ²
1	Shaft fillet	50	-	171,87	60,94	-	-	-	-	-	-
2	Shaft fillet	74	65	143,85	53,68	-	-	-	-	-	-
3	Shaft fillet	94	75,12	163,96	59,05	-	-	-	-	-	-
4	Shaft fillet	125	-	163,96	-	-	-	-	-	-	-
5	Shaft fillet	135	-	143,85	-	-	-	-	-	-	-
6	Calculation results for point x	142	-	295,02	-	-	-	-	-	-	-
Dynamic safety											
No.	Type	Position x mm	S_D		in Point1 S_{D1}		in Point2 S_{D2}				
1	Shaft fillet	50	2,08		-		-				
2	Shaft fillet	74	1,98		-		-				
3	Shaft fillet	94	3,49		-		-				
4	Shaft fillet	125	9,23		-		-				
5	Shaft fillet	135	12,38		-		-				
6	Calculation results for point x	142	1,36e+005		-		-				
Last updated: 13.05.2026					Module version: 18.0.13e			Page: 14 / 31			

Shaft

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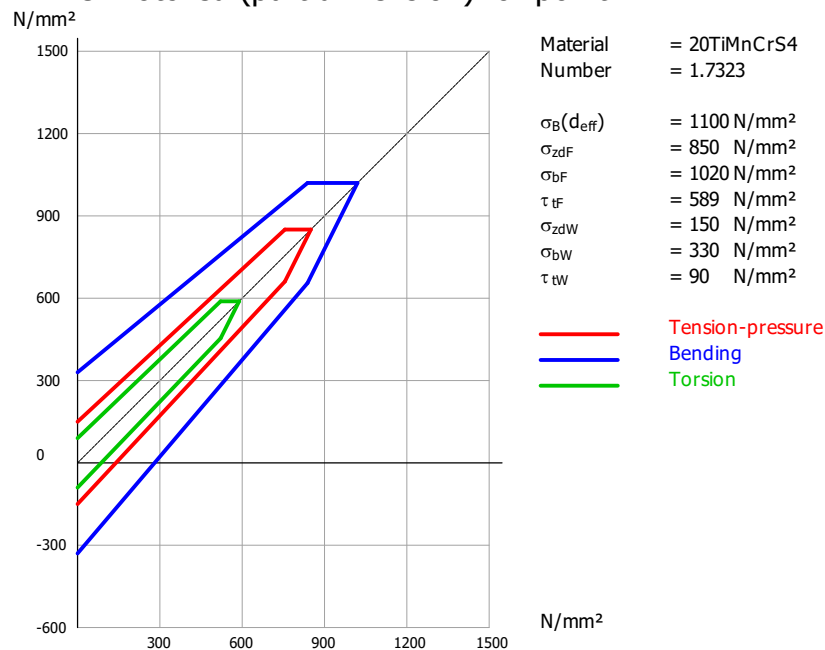
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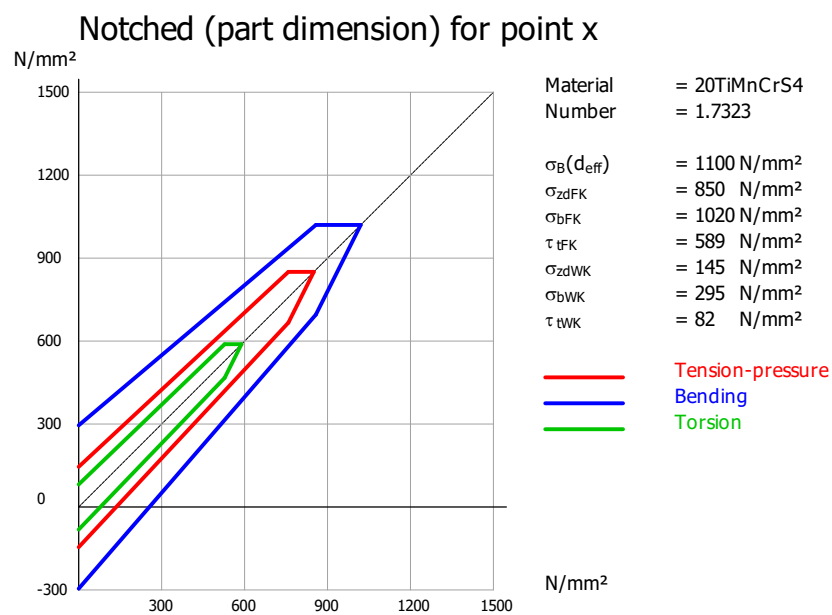
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Safety against incipient crack
with hard surface

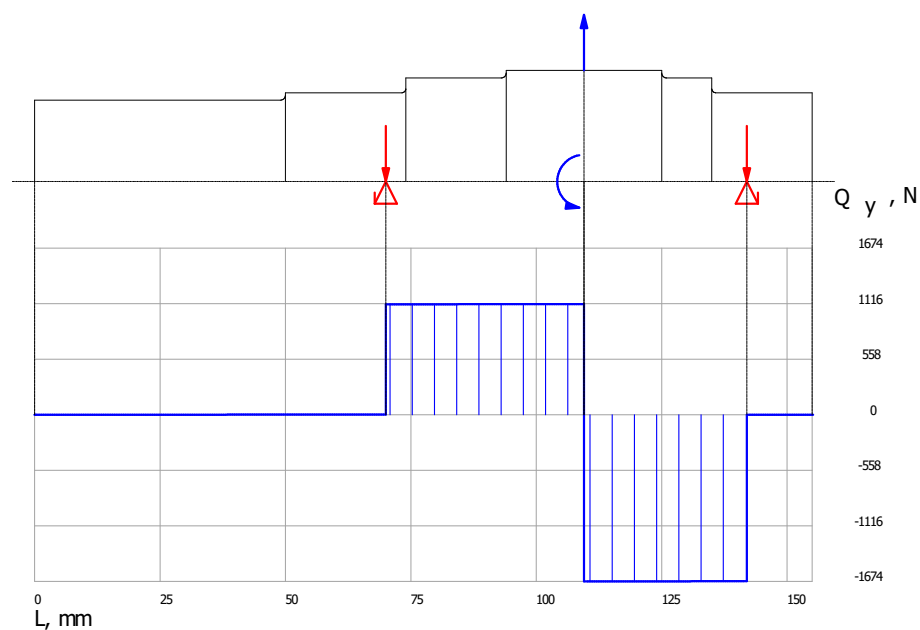
No.	Type	Position x mm	S_G	In Point1 S_{G1}	in Point2 S_{G2}
1	Shaft fillet	50	28,38	-	-
2	Shaft fillet	74	22,3	-	-
3	Shaft fillet	94	40,89	-	-
4	Shaft fillet	125	68,39	-	-
5	Shaft fillet	135	90,14	-	-
6	Calculation results for point x	142	1,06e+006	-	-

Unnotched (part dimension) for point x

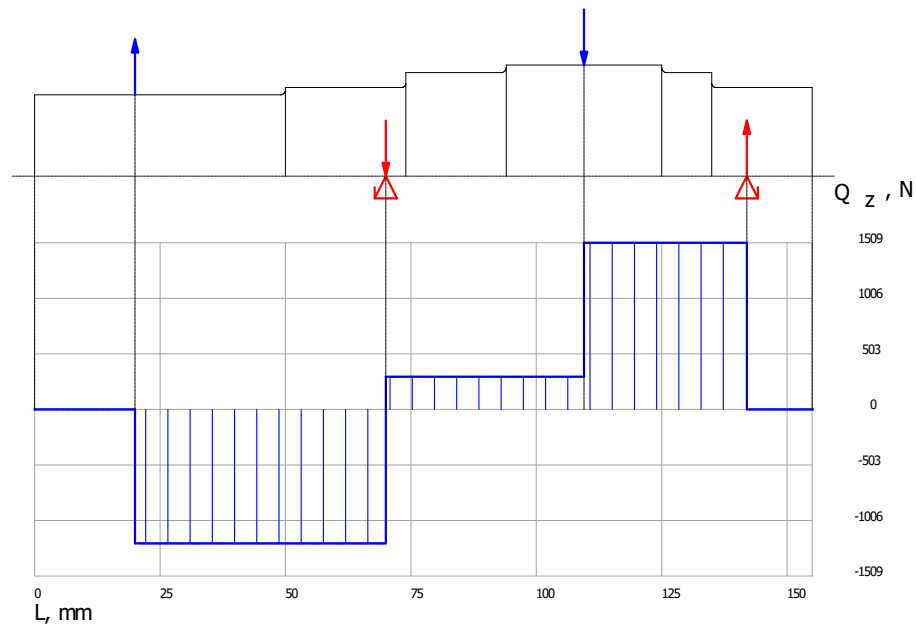




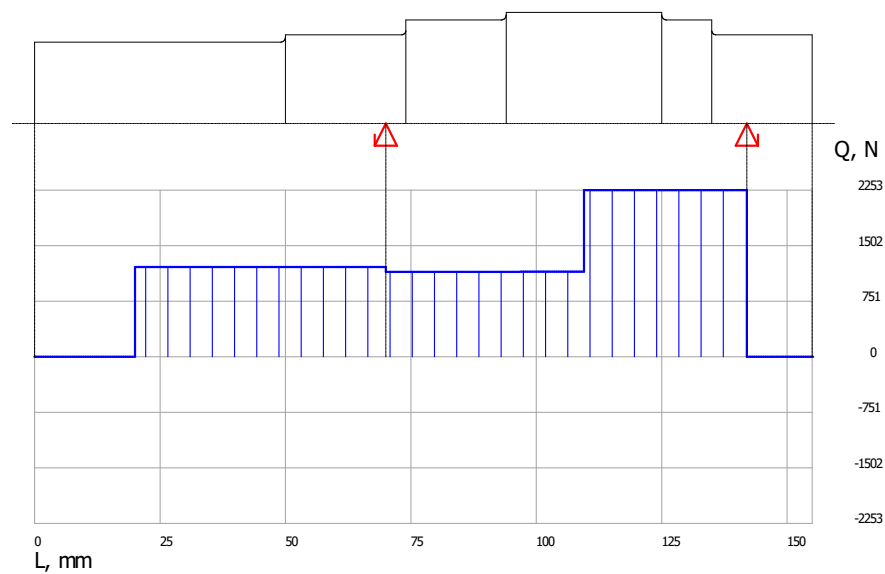
Trend of curve of the transverse force in the Y-X-plane



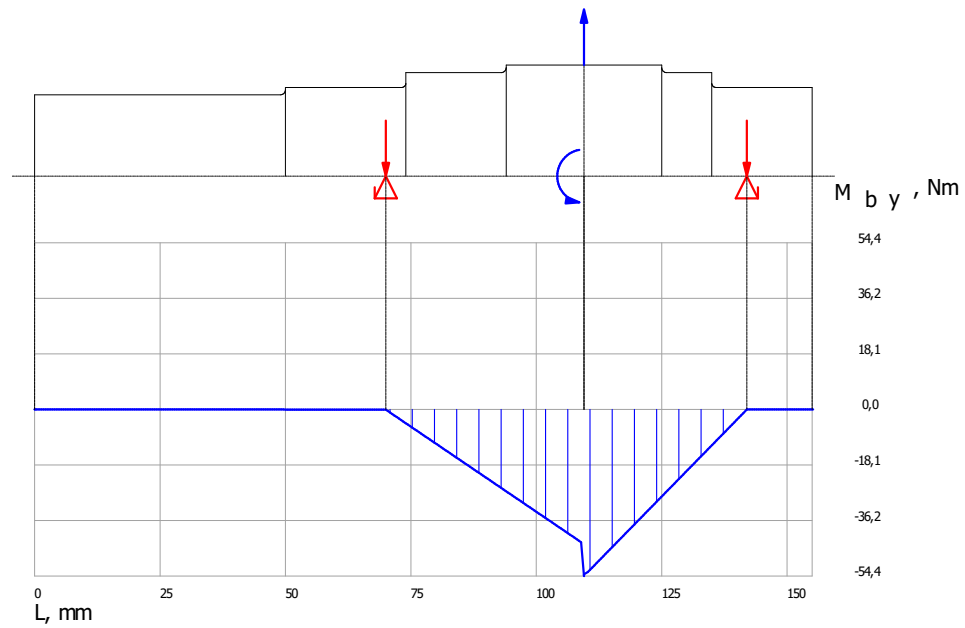
Trend of curve of the transverse force in the Z-X-plane



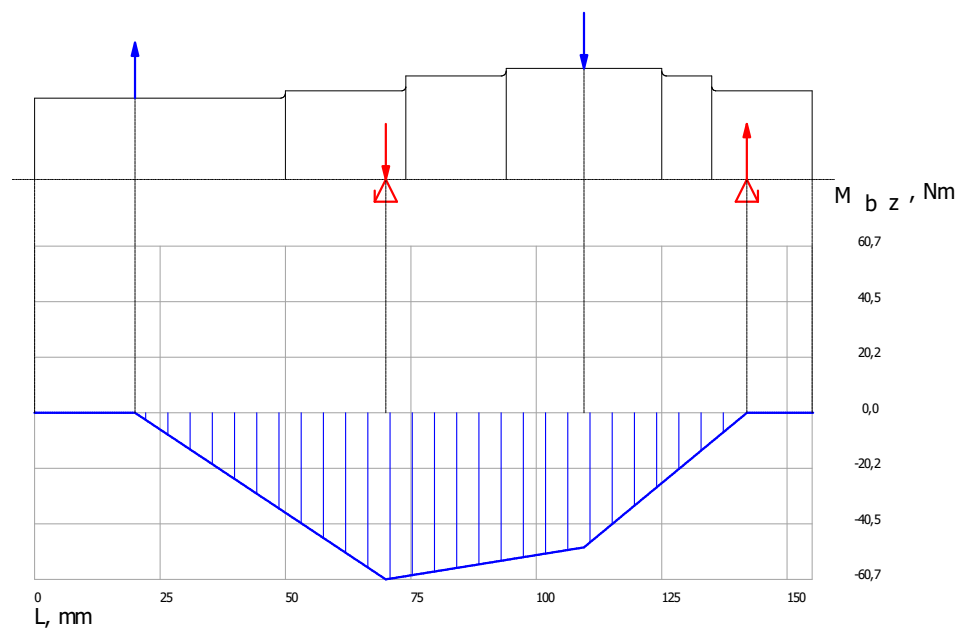
Trend of curve of the transverse force (combined characteristic)



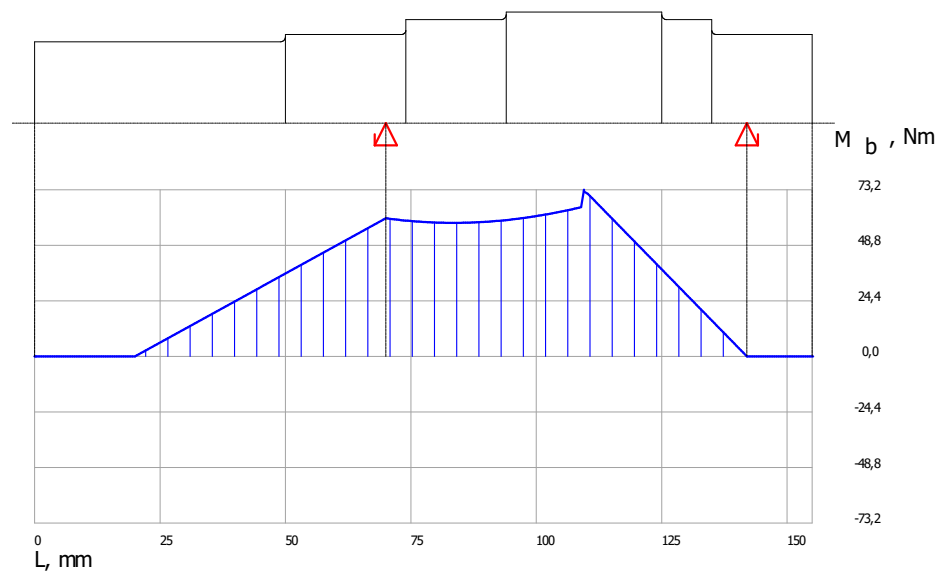
Bending moment in the Y-X-plane



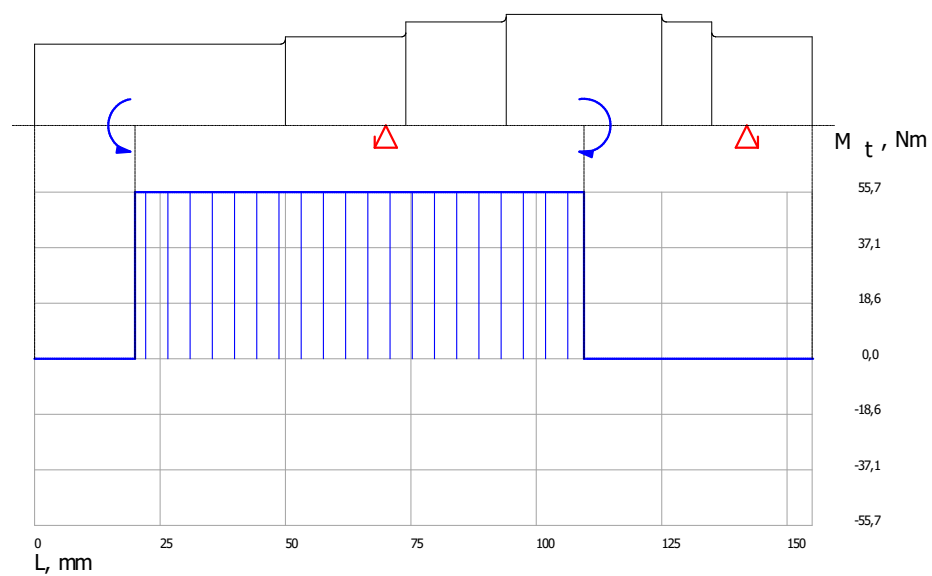
Trend of curve of the bending moment curve in the Z-X plane



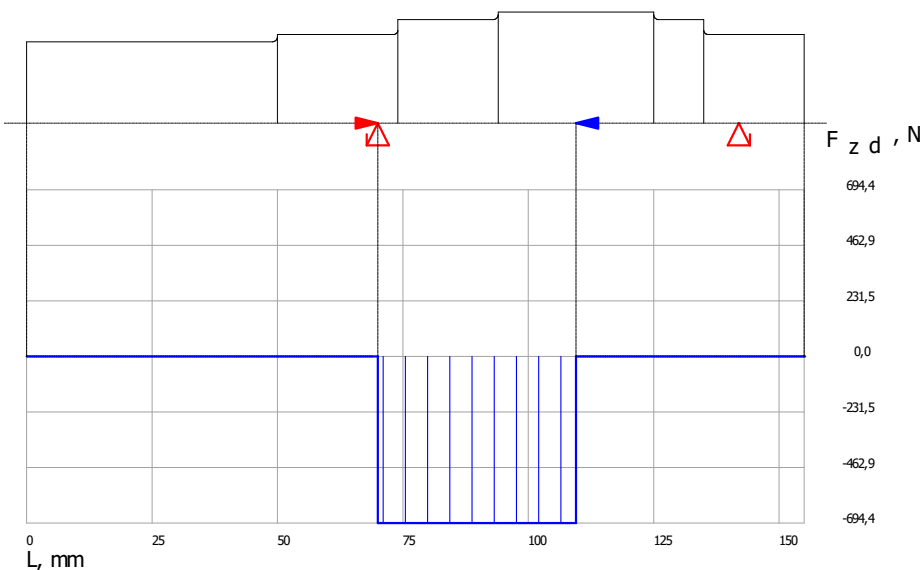
Trend of curve of the bending moment (combined characteristic)



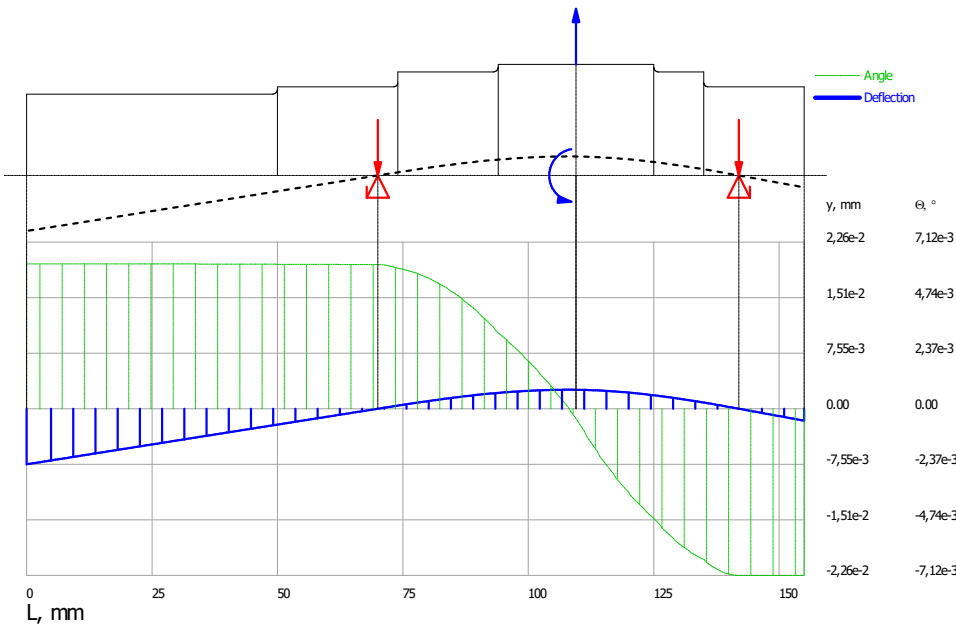
Trend of curve of the torsional moment



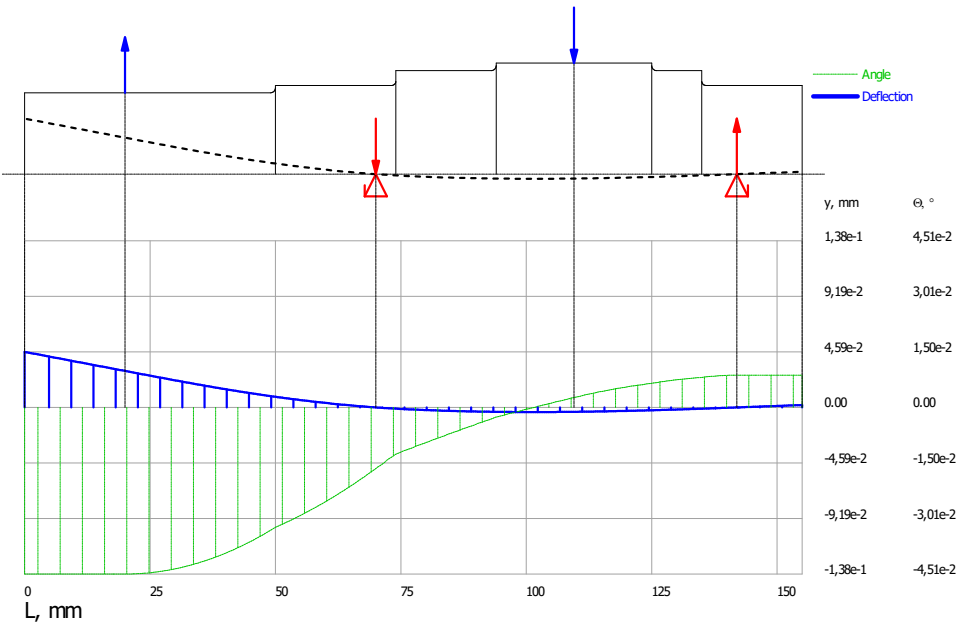
Trend of curve of the tension-pressure forces



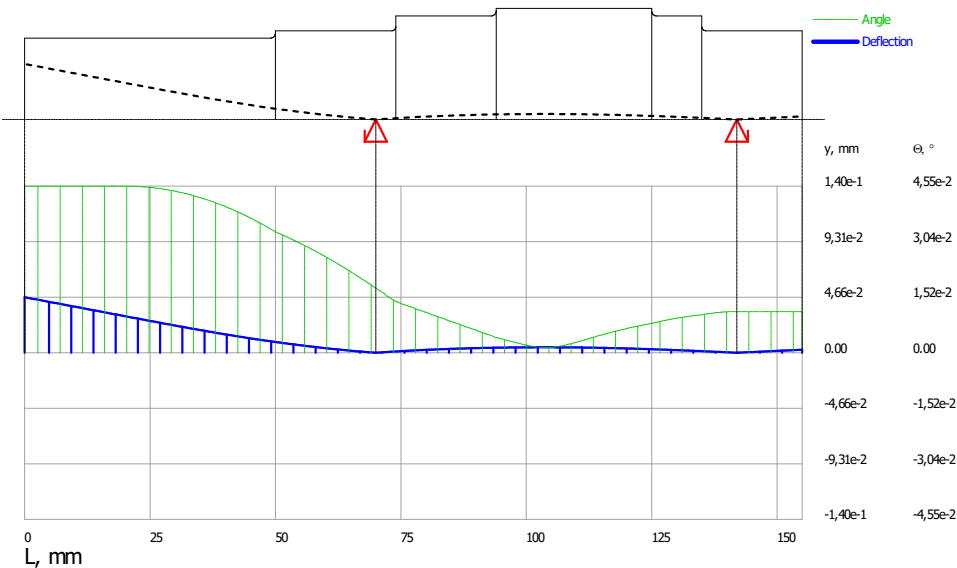
Deflection and angle of deflection in the Y-X-plane



Deflection and angle of deflection in the Z-X-plane



Deflection and angle of deflection (combined characteristic)



Shaft

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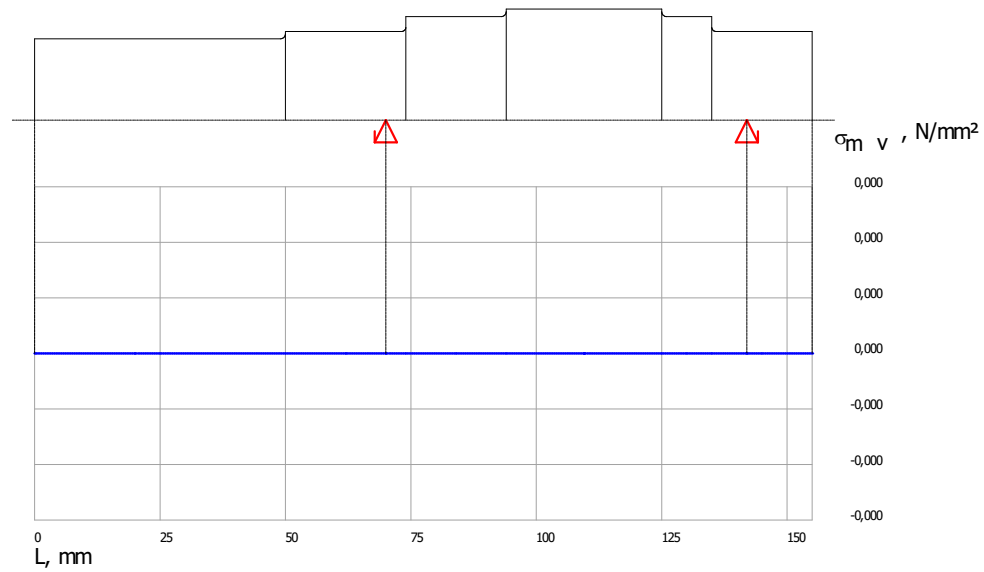
Project:

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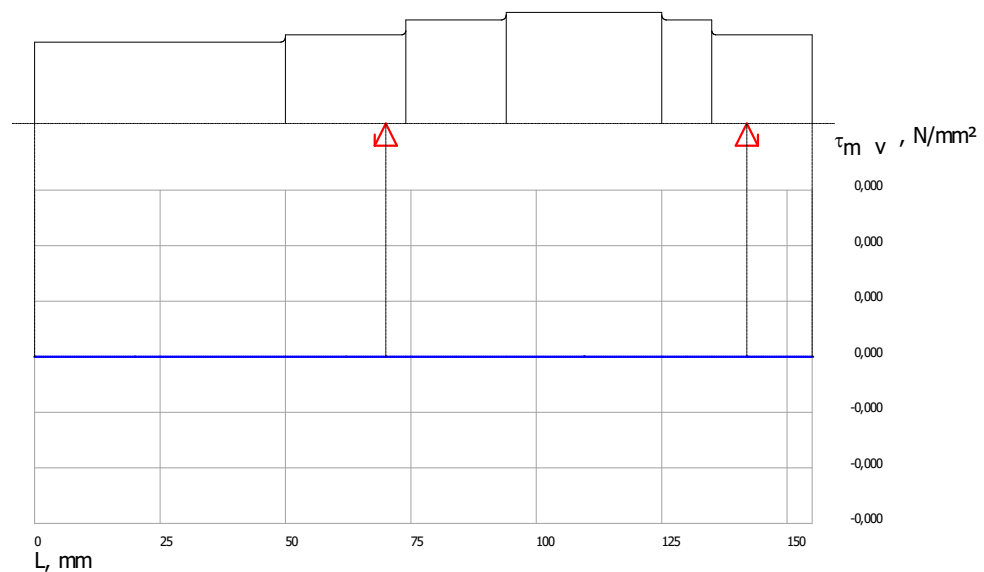
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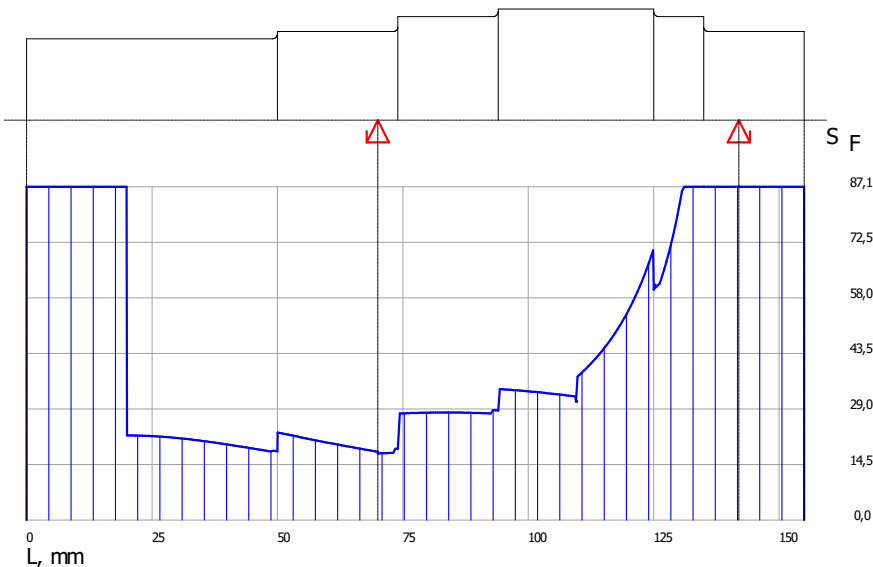
Equivalent mean stress (normal stress)



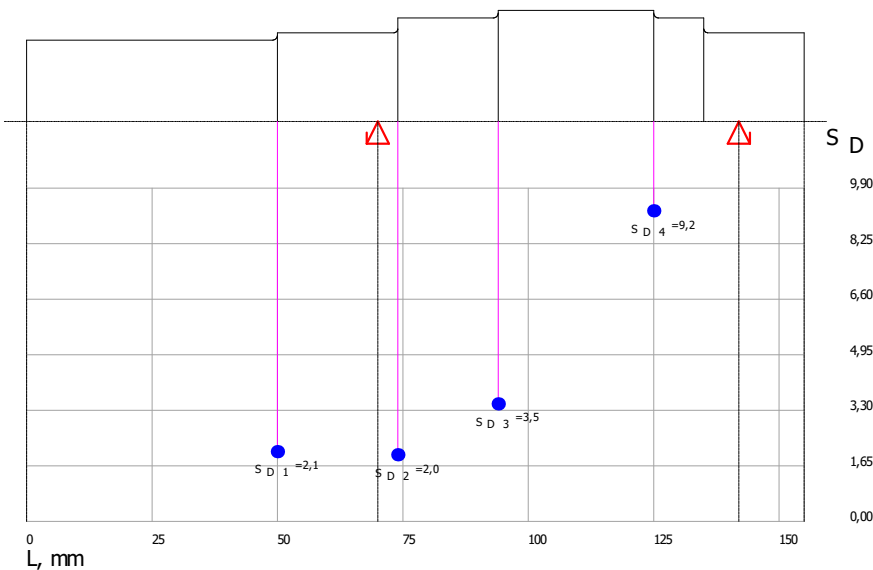
Equivalent mean stress (shear stress)



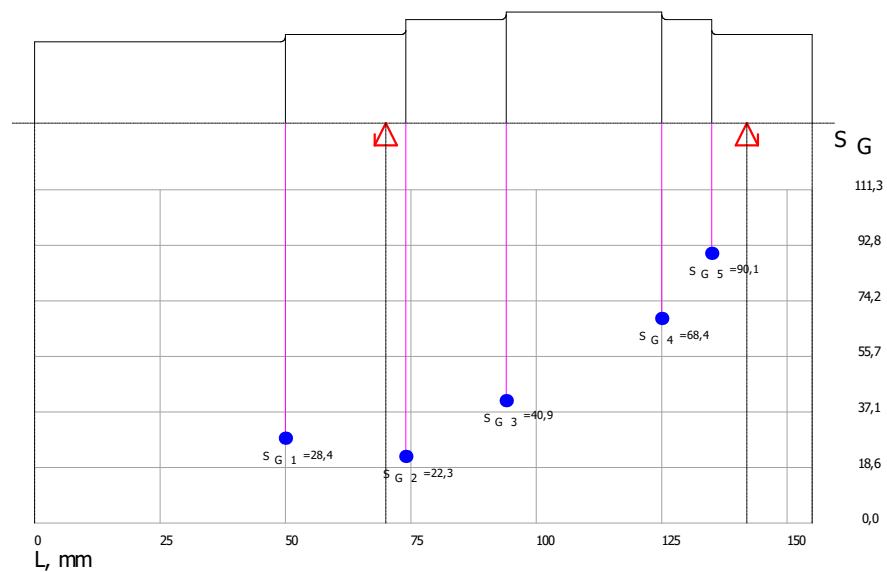
Safety factor against yielding
(diagram section up to 5*minimum safety)



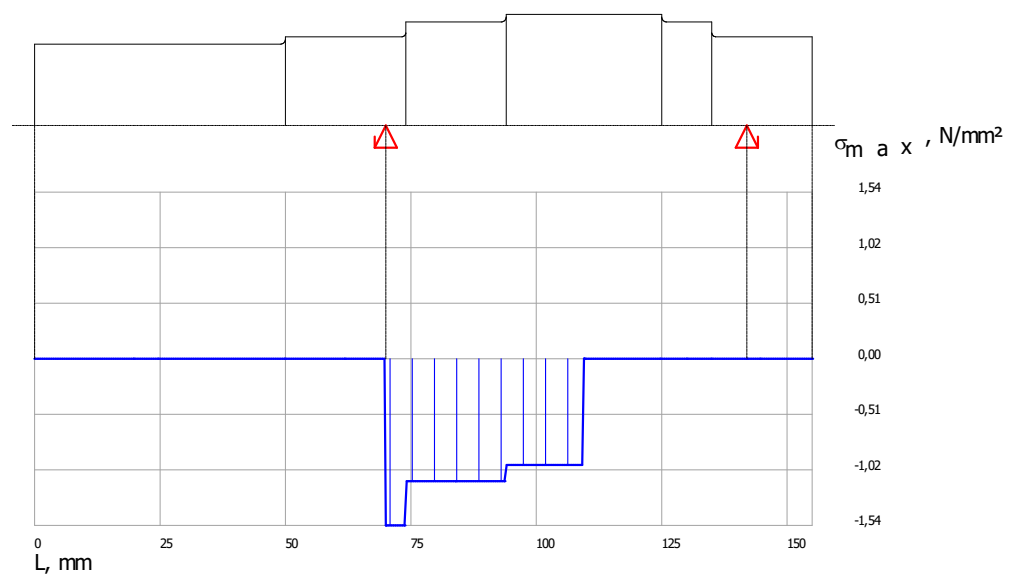
Safety against fatigue fracture
(diagram section up to 5*minimum safety)



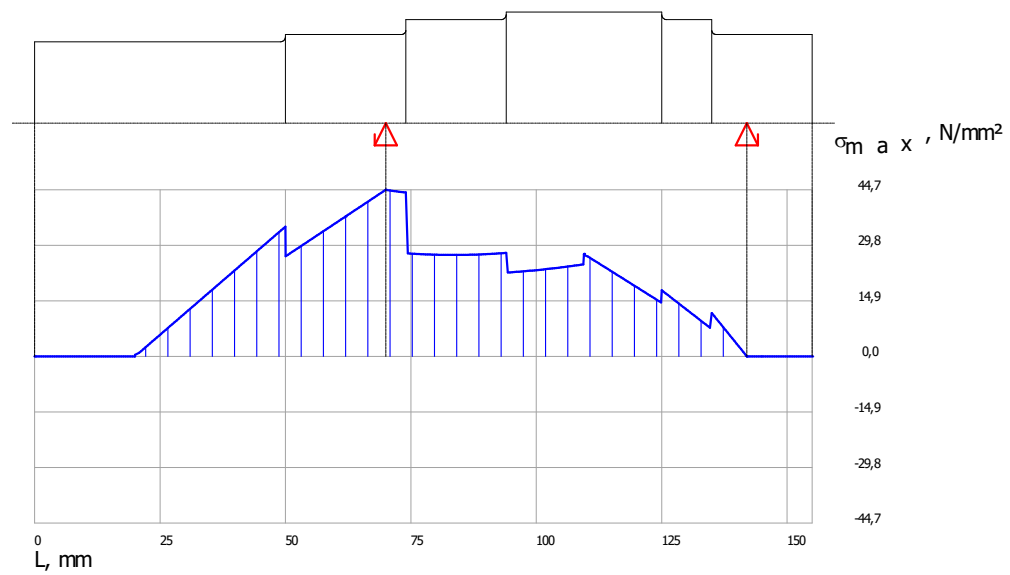
Safety against incipient crack with hard surface
(Cross-section: $S_G = 5 \cdot S_{Gmin}$)



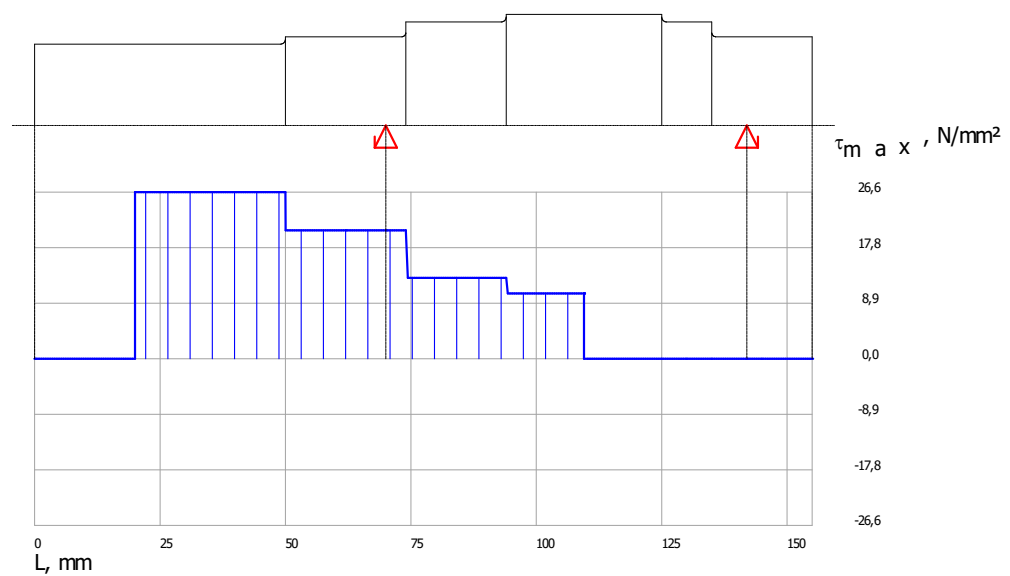
Maximum value of the tension-pressure stress (combined characteristic)



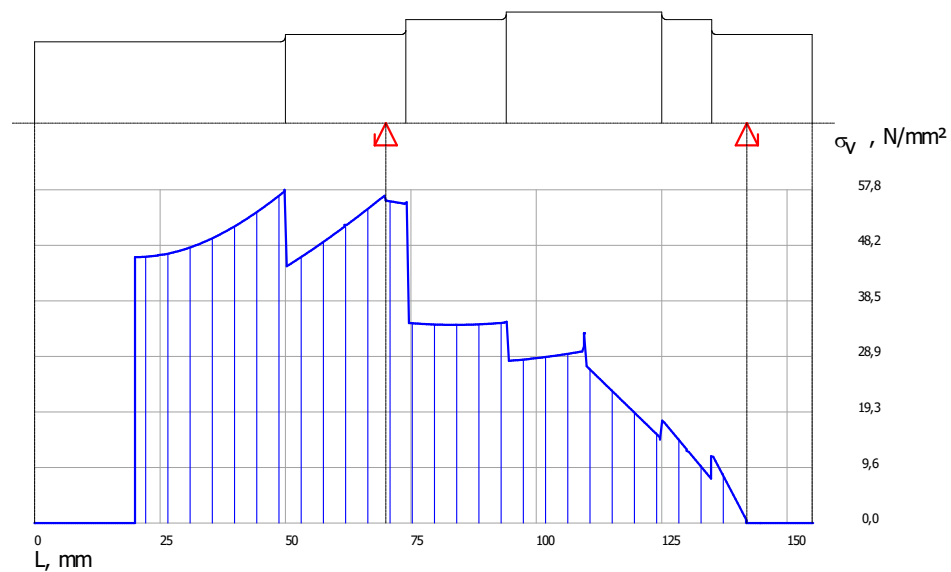
Maximum value of the bending stress (combined characteristic)



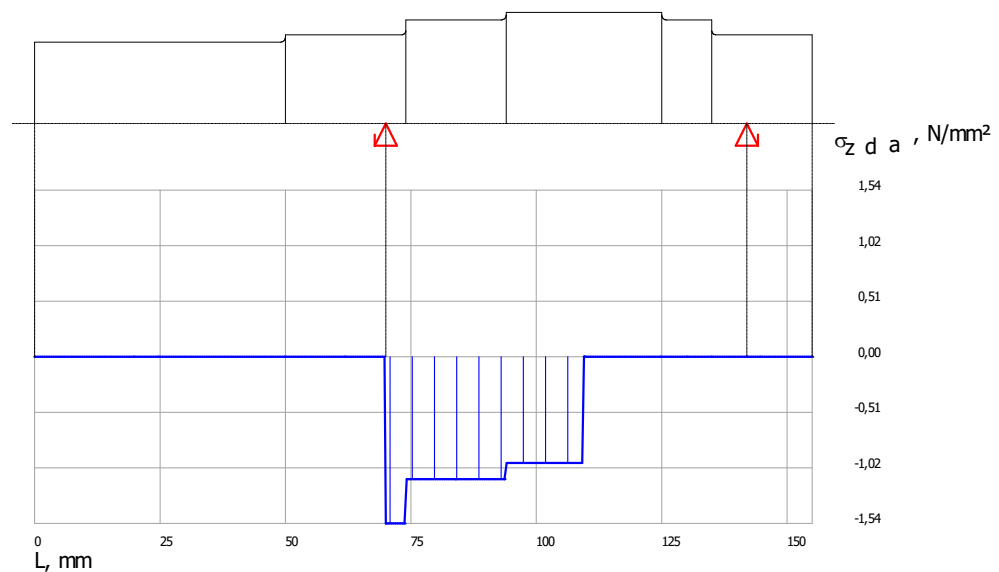
Maximum value of the torsional stress (combined characteristic)



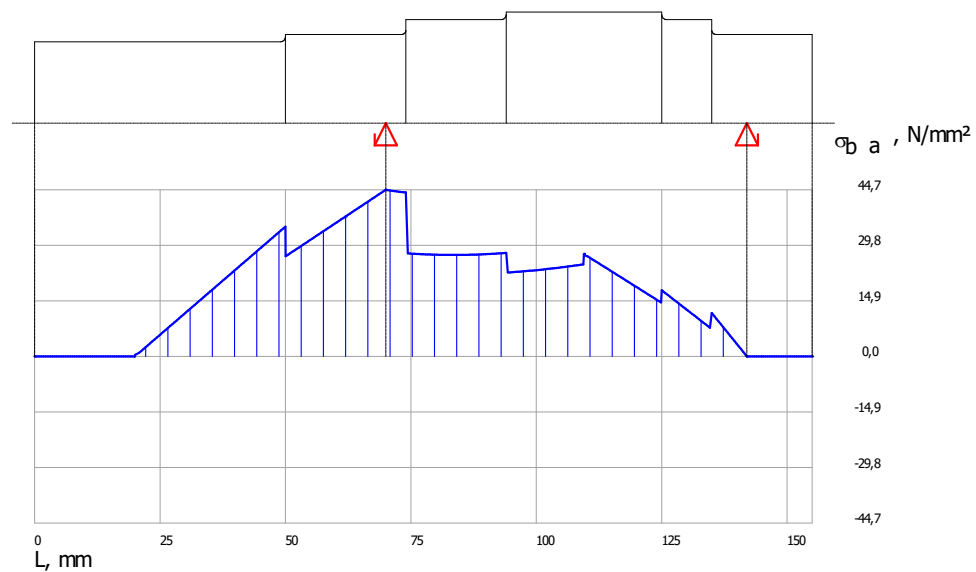
Equivalent stress development (resultant)



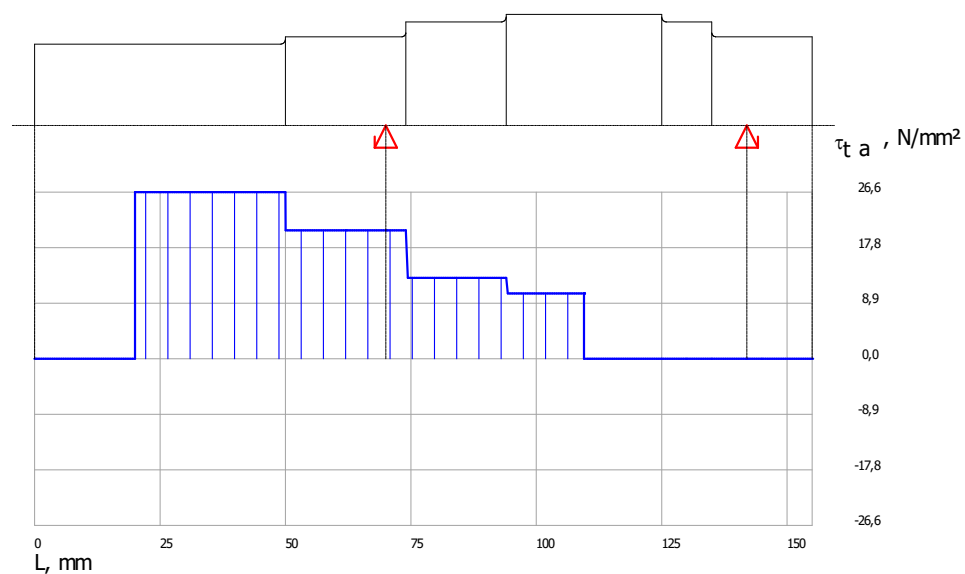
Amplitude value of the tension-pressure stress (combined characteristic)



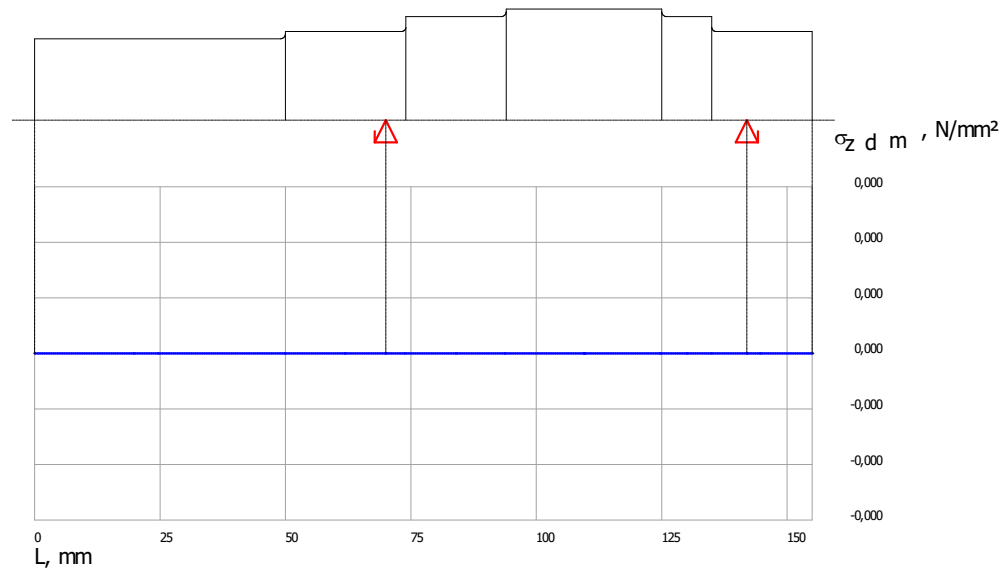
Amplitude value of the bending stress (combined characteristic)



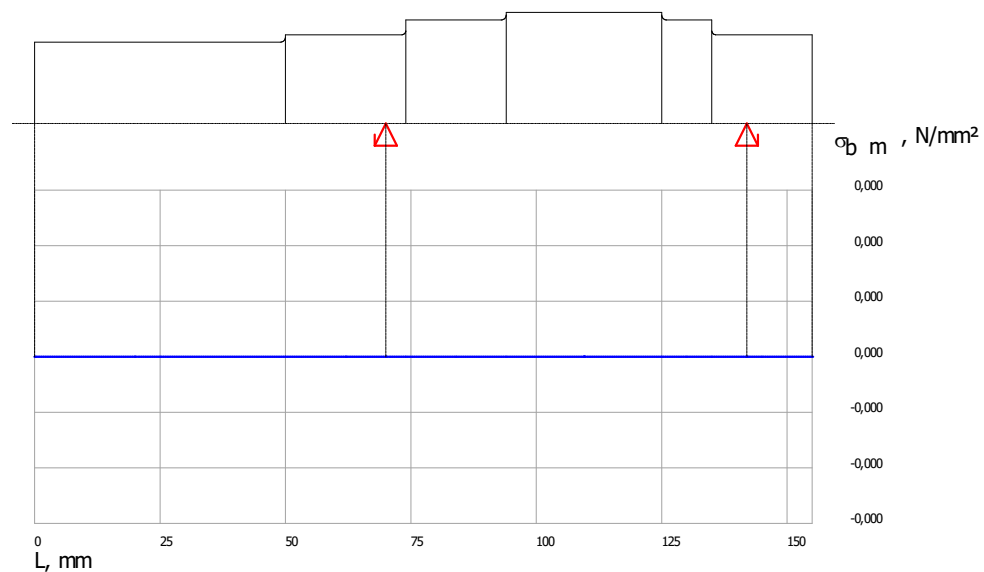
Amplitude value of the torsional stress (combined characteristic)



Mean value of the tension-pressure stress (combined characteristic)



Mean value of the bending stress (combined characteristic)



Shaft



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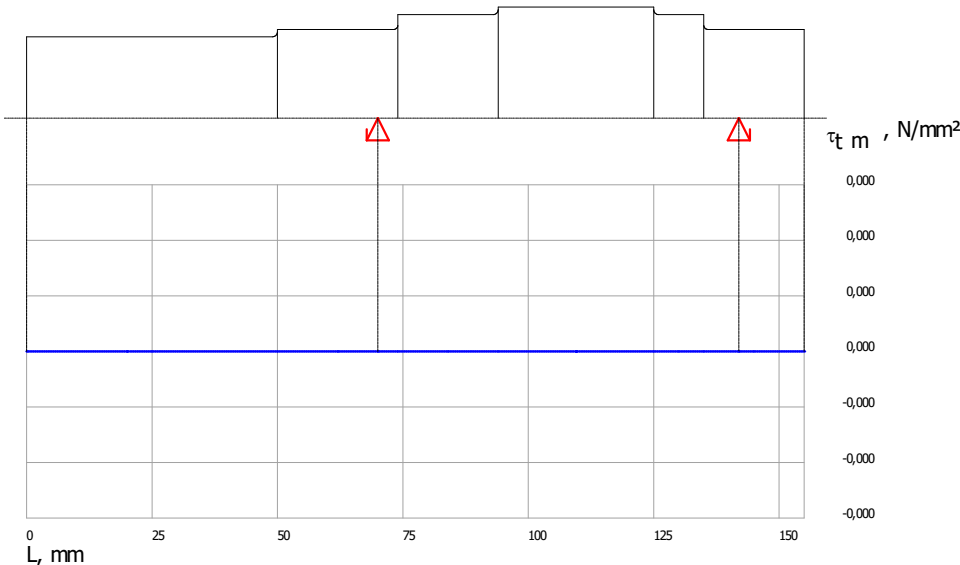
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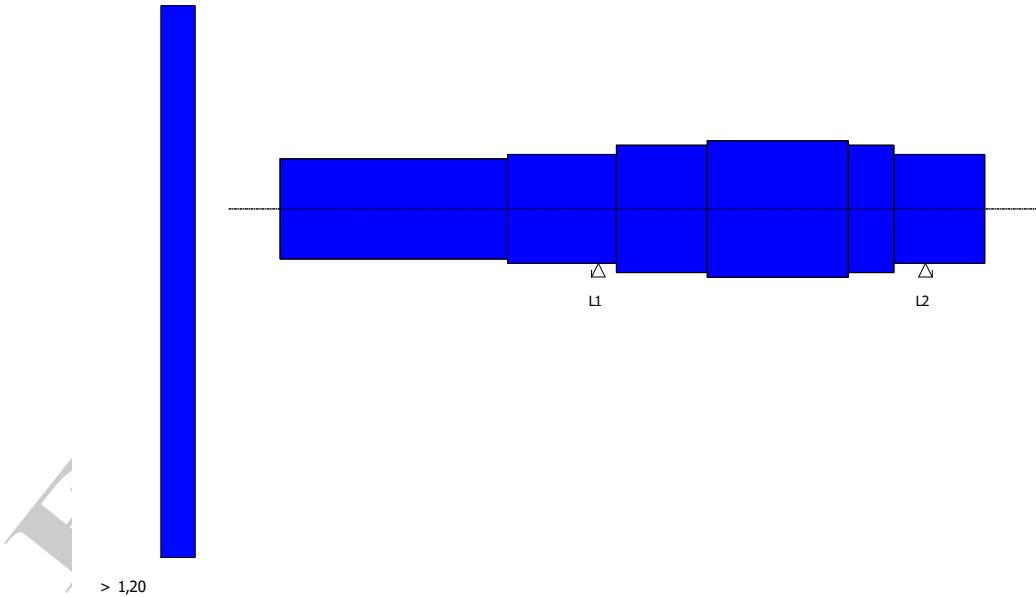
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Mean value of the torsional stress (combined characteristic)

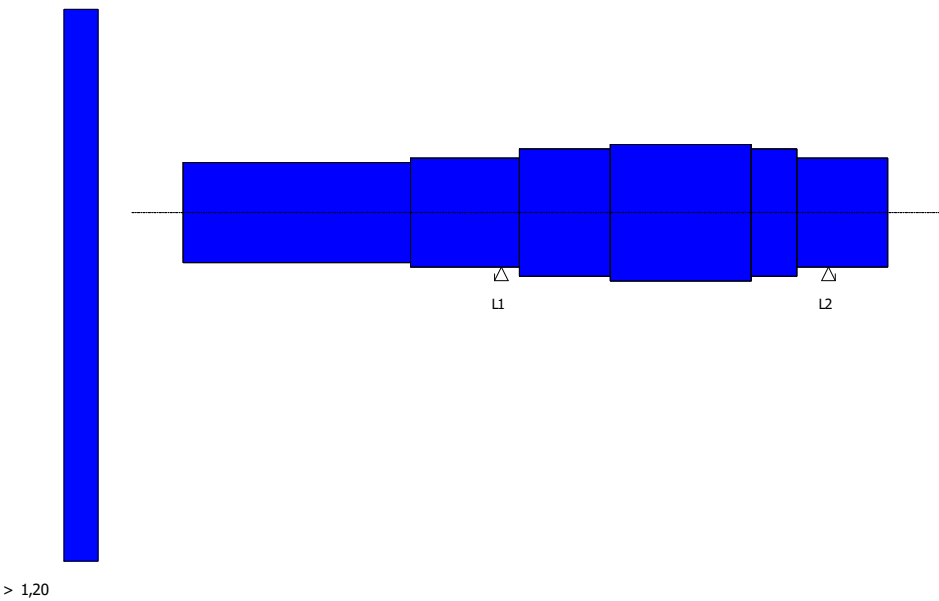


Safety factor against yielding

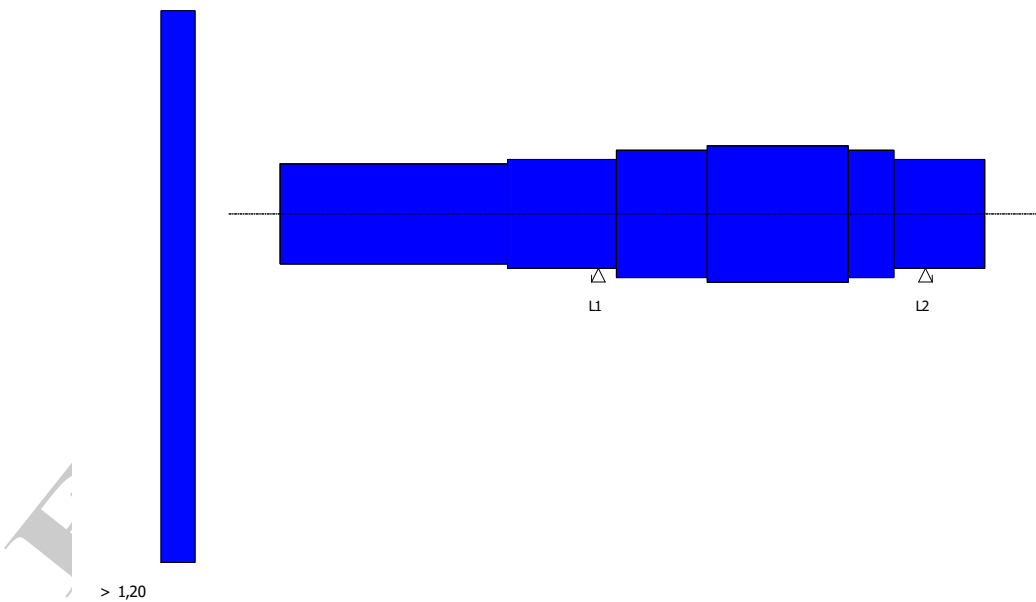


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Safety against fatigue fracture



Safety against incipient crack with hard surface



Shaft

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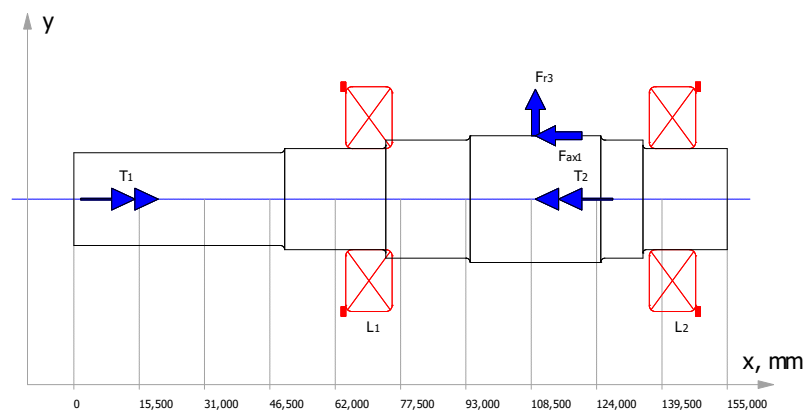
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Calculation graphic Y-X-plane



Calculation graphic Z-X-plane

